Forward Modeling of Scattering Centers From Coated Target on Rough Ground for Remote Sensing Target Recognition Applications

Siyuan He, *Member, IEEE*, Mengbo Hua, Yunhua Zhang, Xiaoyong Du, and Fan Zhang

***Abstract*— This article presents a forward modeling method for the scattering center (SC) model of coated targets on rough ground, tailored for remote sensing and target recognition applications. The methodology initiates with the geometric model and coating impedance parameters of the ground target, employing a forward approach to decompose overall scattering into multiple component-level sources. Integrating high-frequency electromagnetic (EM) theory, quantitative computation, and derivation of SC parameters constructs a parameterized 3-D model for coated targets on rough ground. Illustrated through the example of a T72 tank and considering the effects of coated media and ground coupling, a forward SC model for the tank target on rough ground is developed. The model's statistical characteristics are meticulously analyzed, and its accuracy is validated using real measurement data from publicly available datasets. The proposed model offers several advantages. First, it directly reflects the local geometry and coated material characteristics of target components, providing detailed information about component shapes and materials. Second, the modeling process meticulously considers and segregates the contributions of the target's complex coupled SCs with the ground. A statistical analysis demonstrates consistent parameter convergence, aligning with theoretical expectations. Third, this research extends the forward modeling approach to coated targets on rough ground, enabling the model not only for component-level identification but also for rapid target synthetic aperture radar (SAR) echo reconstruction. In conclusion, this cost-effective approach provides a practical means for the rapid estimation and analysis of radar characteristics, target detection and recognition, target remote sensing, and other applications in complex environments.**

***Index Terms*— Coating target, forward modeling of scattering centers (SCs), remote sensing and recognition applications, rough ground, scattering characteristics.**

I. Introduction

THE target scattering center (SC) model offers a concise and rational description of the electromagnetic (EM) scattering characteristics of target, making it widely utilized in SAR automatic target recognition (SAR ATR) systems. Derived from the underlying physical mechanisms, the target SC represents the EM scattering of target feature structures, encapsulating rich information about the target's structure. Modeling and analyzing SCs play a pivotal role in radar target identification and interpretation and have even become a critical component in the process of remote sensing identification and interpretation [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18].

The approaches to SC modeling can be broadly categorized into forward and inverse methods. Traditional inverse approaches involve parameter estimation by using measured or simulated radar echo data to retrieve the SC parameters. The inherent uncertainty in the inverse approach leads to models lacking clear physical interpretations and insufficient target feature information [1], [2], [3], [4], [5], [6], [7], [8], [9]. Researchers from the Electromagnetic Engineering Laboratory (EEL) at Wuhan University [10], [11] have proposed a forward modeling method for complex target component-level SCs based on a geometric model. This method uses high-frequency theory to forward separate and quantitatively compute the scattering contributions of the SCs. The model parameters are derived based on EM mechanisms and target geometry, introducing physical constraints to eliminate the multivaluedness of the scattering sources. This enhances the model's determinism and establishes a well-corresponded parametric model and its parameters with the components and local structures of the target. This method has the capability to concurrently provide comprehensive target scattering data and scattering data for various target components, along with the concealed information concerning the components and structural characteristics of the target hidden behind them. The parameter values of the SCs are thereby employed as a representation of this information at the EM scattering level. Subsequently, the SCs forward modeling method is closely integrated with SAR target recognition, leading to the proposal of an SAR target rapid indexing method based on geometric model contour region features. A hierarchical SAR ATR system is designed, leveraging the forward modeling of SCs [12].

The forward modeling method of SCs has garnered widespread attention in the field of target recognition, receiving numerous citations and applications [13], [14], [15], [16], [17], [18]. Jackson [13], renowned international scholar, has emphasized the significant application value of the high-frequency ray-based forward modeling approach in SAR target

Manuscript received 28 July 2023; revised 17 October 2023 and 13 November 2023; accepted 12 December 2023. Date of publication 18 December 2023; date of current version 29 December 2023. This work was supported by the National Natural Science Foundation of China under Grant 62231026 and Grant 62301215. *(Corresponding author: Siyuan He.)*

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Digital Object Identifier 10.1109/TGRS.2023.3344128

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recognition. As indicated in [5] and [6], the SC model obtained through forward modeling succinctly encapsulates the scattering mechanisms of typical scatterers, offering rich information about the geometric shape of the targets. These models are successfully applied in SAR ATR and EM inverse modeling. Ding and Wen [14], Ding et al. [15], Ding and Wen [16], and Yang et al. [17], [18] fully utilized the advantages of high accuracy, local separability, stable recognition features, simplicity in calculations, and strong physical interpretability of the forward EM model. Various matching criteria are proposed based on SC positions and attribute parameters. SAR target recognition methods have been designed, driven by the forward EM scattering model, and it has been demonstrated that this approach exhibits strong adaptability and effectively supports the improvement of recognition accuracy and target identification performance under standard operating conditions.

Despite recent progress in the forward modeling of SCs, challenges persist in practical recognition. Real-world targets, differing from ideal perfect electric conductors (PECs), frequently incorporate nonmetallic dielectric materials such as absorbent coatings or antirust paints on high-conductivity alloys. This necessitates a study of changes in EM scattering characteristics and variation in strong SCs induced by coatings. Furthermore, the intricate EM coupling between targets and the ground significantly impacts target attributes. Accounting for coupling scattering is imperative. Broadening the scope of SC forward modeling to encompass coated targets in complex real-world environments is essential. Such an approach ensures that SC models hold genuine physical significance for practical scenarios and effectively support applications such as remote sensing target recognition.

In response to these challenges, a forward modeling method for SCs of coated targets on rough ground, tailored specifically for remote sensing and recognition applications, is proposed. The methodology commences with the consideration of the geometric model of the ground target and the impedance parameters of the partitioned coating materials. Through the integration of spatial ray tracing and diversity techniques, the overall scattering characteristics of the target are decomposed and discretized into multiple component-level scattering sources. These sources encompass individual scattering sources associated with the target and coupled scattering sources existing between the target and the ground. Subsequently, utilizing high-frequency EM theory, the quantitative calculation and derivation of the 3-D phase center (position parameters), scattering contributions (amplitude parameters), frequency-dependent factors, and distribution widths of the SCs are performed. This process results in the forward construction of a parametric model for the 3-D SCs of coated targets on rough ground.

This research utilizes the T72 tank from the moving and stationary target acquisition and recognition (MSTAR) database as a case example. Initially, the strong SC distribution for the target is obtained through forward SC modeling for a PEC target. Subsequently, absorptive materials are applied to coat strong scattering structures on the target, such as the turret and wheelhouse, corresponding to strong SCs. SC model parameters for the coated target are calculated in a forward manner, constructing a model that establishes the correlation between target geometry and coating materials. Experimental results effectively suppress strong SC amplitudes for the coated target, altering the distribution and relative amplitude of strong scattering sources. The forward model for the coated target significantly outperforms the PEC scenario in SAR image reconstruction but faces the challenge of missing coupled SCs. To address this, rough ground is introduced to analyze ground-to-target coupled scattering. A forward model for a coated target on rough ground is constructed, and statistical analysis refines the modeling process. This ensures that the results more accurately reflect the true scattering characteristics, accurately considering and separating contributions of complex coupled SCs. Statistical results demonstrate the convergence of the coupled SC parameters, entirely consistent with theoretical expectations.

This research extends forward modeling to targets with dielectric coatings in real rough ground environments. The coupling of SCs between the target and the rough surface significantly increases the similarity between reconstructed SAR images and measured data to over 83%, with a position matching rate of 91.95%, validated using publicly available dataset measurements. A comparative experiment with the SC inverse extraction method based on simulated data enhances research comprehensiveness. Unlike the inverse method, the forward modeling method establishes a clear SC-local structure correspondence, eliminating complex iterative optimizations. The model parameters carry rich physical meaning. The forward model excels in providing a clear physical deconstruction of the target's EM scattering, facilitating in-depth exploration of target feature matching information. The main advantages of the proposed method are as follows.

1) Model parameters effectively capture detailed local target geometry and coating material characteristics, facilitating remote sensing target recognition by providing information on component shapes and materials.
2) The modeling process accurately incorporates and distinguishes complex coupling of SCs between the target and the ground. Statistical results validate the convergence of coupling SC parameters, consistent with theoretical expectations.
3) This research extends forward modeling beyond PEC targets in free space to practical coated targets in rough ground environments. The SC model offers component-level insights for recognition and enables rapid reconstruction of target SAR echoes.

Compared to traditional inverse optimization methods based on echo data for SC retrieval, the proposed method follows a typical "cause-to-effect" forward approach, providing a clear physical interpretation and understanding of the scattering mechanisms. This research offers a cost-effective means for the rapid estimation and analysis of radar characteristics in complex real-world environments, target detection and recognition, and target remote sensing applications.

The remainder of this article is organized as follows. Section II introduces the forward modeling method for SCs of complex coated targets on rough ground, covering the forward

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calculation of model parameters and statistical analysis of coupled SCs. This section presents the overall framework of this method. Section III includes case analyses, starting with a simplified tank for scattering source analysis and absorptive coating for radar cross section (RCS) reduction. Then, using the T72 tank as an example, the forward SC model is established at multiple angles, and the impact of coating materials and rough ground on the SCs is analyzed. Two similarity evaluation metrics, the normalized similarity index and position matching rate, assess the method's effectiveness and robustness. Section III also conducts comparative experiments with the SC inverse extraction method based on simulated data to highlight the forward modeling method's advantages. Section IV concludes this article with further discussions.

II. Forward Modeling of SCs for Complex Coated Target on Rough Ground

This section outlines the forward modeling approach for SCs of complex coated targets situated on rough ground. It primarily focuses on the forward computation of model parameters and the analysis of statistics pertaining to coupled SCs.

The forward SC modeling method begins by dissecting the geometric model of intricate targets, dividing them into multiple constituent components, and further subdividing each component's surface for distinction. Spatial ray tracing and diversity techniques, based on mixed geometrical optics–physical optics (GO–PO), are then applied to ascertain the trajectories of all rays in space. These rays are categorized based on their paths, segmenting the scattering field from the target into multiple ray subsets. The GO–PO algorithm is also employed to compute the EM scattering field carried by each ray. Each scattering source's contribution results from superimposing the EM scattering fields of all rays within its subset. The strong SCs, having dominant roles due to their significant scattering field magnitudes, are chosen as the strong SCs at the present angle. Accounting for the target's intricate scattering characteristics and mechanisms, the SCs are expressed as an attribute SC model. The specific expression is as follows:

$$\begin{aligned} &E_i^s(f,\theta,\varphi,r_i) \\ &= A_i\left(j\frac{f}{f_c}\right)^{\alpha_i}\cdot \sin c\left[\frac{2\pi f}{c}L_i\sin\left(\varphi-\overline{\varphi}_i\right)\right] \\ &\quad\cdot\exp\left[\frac{-4j\pi f}{c}(x_i\sin\theta\cos\varphi+y_i\sin\theta\sin\varphi+z_i\cos\theta)\right] \\ &\quad\cdot\exp(-2\pi f\gamma_i\sin\varphi) \end{aligned} \tag{1}$$

where A_i is the amplitude parameter of the ith SC, f is the frequency of the incident wave, f_c represents the SC frequency, c is the speed of light, L_i is the length parameter of the SC, α_i denotes the frequency dependence, $r_i=[x_i, y_i, z_i]$ is the 3-D position vector of the SC, φ is the aspect angle, and $\overline{\varphi}_i$ is the orientation angle of the distributed SC. $L_i=0$ if the SC is local, and otherwise, $\gamma_i=0$.

The scattered field of the target can be represented as the sum of the contributions from N SCs, which can be expressed as

$$E_{\text{total}}^s(f,\phi)=\sum_{i=1}^{N}E_i^s(f,\theta,\varphi,r_i). \tag{2}$$

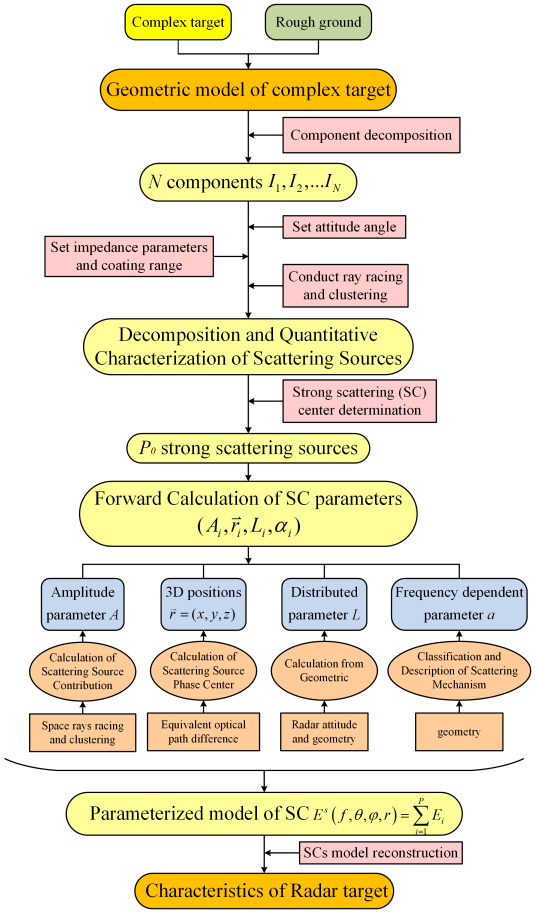


Fig. 1. Flowchart for parametric forward modeling of SCs for coated targets on rough ground.

SC forward modeling exclusively involves computing the target's SC parameter set for a given observation position. By applying a narrow range of azimuth angle variations and corresponding step sizes, along with a frequency range with associated increments, and incorporating these values into (1), the SAR image of the target is reconstructed. This approach strikes a balance between precision and significantly reduced data demands and computational time compared to measurement or simulation techniques. It enables rapid broadband radar echo simulations and real-time complex scattering simulations.

The flowchart depicting the parametric forward modeling of SCs for coated targets on rough ground is presented in Fig. 1.

Compared to ideal conductive targets, the impact introduced by thin coatings primarily manifests as an attenuation effect on the amplitude parameter A of the scattering field [19]. Parameters, such as the 3-D phase center of the scattering source, frequency-dependent factors, and distribution width, are directly calculated from the target's geometric structure.

In this research, the amplitude parameter A is determined using the mixed GO–PO high-frequency algorithm based on the impedance boundary condition (IBC). The field amplitude

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on the primary ray is directly calculated using the PO algorithm, while for field amplitudes on secondary and higher order coupled rays, the mixed GO–PO approach is employed [20], [21].

The length parameter L primarily characterizes the scattering characteristics of distributed SCs exhibiting "coherent" scattering features, which manifest as noticeable azimuthal "length" characteristics in radar SAR images. Therefore, as long as two equivalent endpoints of a "bright line" are found within the set of rays associated with distributed SCs, the length parameter of that SC can be obtained by projecting and subtracting the two endpoints

$$L_i = |(x_m - x_n)\sin\theta\sin\varphi - (y_m - y_n)\sin\theta\cos\varphi| \tag{3}$$

where (x_m, y_m) and (x_n, y_n) are the coordinate positions of the two endpoints, and θ and φ are the zenith angle and azimuth angle of the radar incident wave, respectively.

The frequency-dependent factor α describes the extent to which the scattering intensity of the SC is related to the frequency. This parameter can be determined by examining the target's geometric information and assessing the scattering mechanisms contained within each SC [11].

The 3-D positions of SCs (equivalent interaction position of rays) can be determined by summing up the weighted reflection points of all rays within the respective subset. These positions determine the phase of the scattering electric field generated as the incident rays travel from the radar to the interaction point. For single reflection rays, the reflection point is taken as the geometric center of the illuminated area (the SC position). For multiple reflection rays, the principle of calculating phase difference through equivalent path length difference is applied to obtain the reflection points [22]. The weight values are determined by the magnitude of the induced currents on the target [10].

A. Forward Calculation of Amplitude Parameter

The amplitude parameter A is determined through the use of the IBC-based mixed GO–PO high-frequency algorithm. The IBC establishes a link between tangential electric and magnetic field vectors and the equivalent surface impedance of dielectric-coated targets, simplifying the computation of EM scattering for thin-coated dielectric targets without considering internal field distribution [23], [24], [25]. This article employs the IBC-based mixed GO–PO algorithm to calculate direct scattering between surface elements of electrically large dielectric-coated targets and multiple coupling scattering contributions between these elements. This is combined with the SC forward modeling method from [10] to deduce model parameters for coated targets.

For thin-coated and electrically large targets, the reflection from the dielectric layer quantitatively characterizes primary scattering characteristics. The subsequent sections discuss the reflection characteristics of impedance surfaces. Coatings with anisotropic absorbing materials dissipate radar incident wave energy and can scatter EM waves in various directions by controlling the principal axis direction of anisotropic materials. The anisotropic medium model can be simplified to an isotropic medium model. This article primarily focuses on the scattering of anisotropic impedance surfaces.

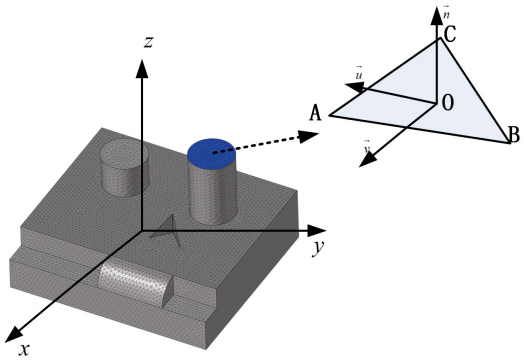


Fig. 2. Schematic of the target's impedance surface.

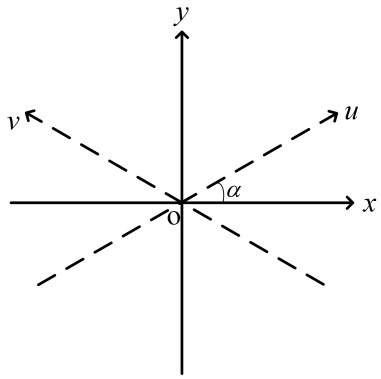


Fig. 3. Local reference coordinates on the target surface and the directions of anisotropic optical axes.

Fig. 2 shows a schematic of the target's impedance surface, whose reflection coefficient can be expressed as follows:

$$\overline{\overline{R}} = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}. \tag{4}$$

The relationship between the reflected field and the incident field is as follows:

$$\vec{E}^r = \begin{pmatrix} E^r_{//} \\ E^r_{\perp} \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix} \begin{pmatrix} E^i_{//} \\ E^i_{\perp} \end{pmatrix}. \tag{5}$$

On the target surface, a local reference coordinate system (x, y, z) is established, where $\hat{x}$ and $\hat{y}$ are the unit tangential vectors of the surface and $\hat{z}$ is the outer normal unit vector. The unit vectors $\hat{u}$ and $\hat{v}$ are used to represent the orientation of the optical axis of the anisotropic medium, as shown in Fig. 3. The angle between the u-axis and the x-axis is denoted by α, which is referred to as the optical axis declination angle.

The vectorial relation between the impedance plane and the incident wave is shown in Fig. 4. $\hat{\tau}$ is the unit vector of the projection of the incident wave direction in the impedance plane, i.e., $\hat{\tau} = \hat{n} \times \hat{e}_{\perp}$. The normal direction of the plane coincides with the z-axis, i.e., $\hat{n} = \hat{z}$.

The surface impedance of the plane can be represented as follows:

$$\overline{\overline{Z}}_s = Z_{xx}\hat{x}\hat{x} + Z_{xy}\hat{x}\hat{y} + Z_{yx}\hat{y}\hat{x} + Z_{yy}\hat{y}\hat{y}. \tag{6}$$

According to the Leontovich IBC $\hat{n}\times(\hat{n}\times\vec{E}) = -\overline{\overline{Z}}_s(\hat{n}\times\vec{H})$, the components of the total field in the x- and y-directions are

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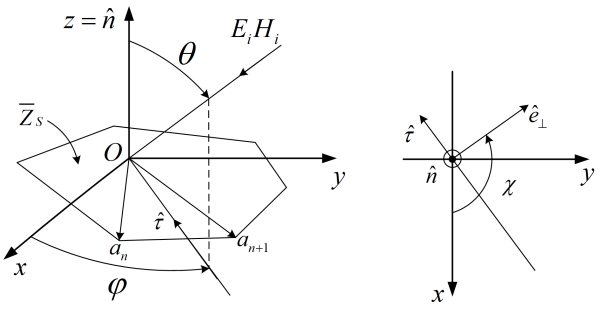


Fig. 4. Vectorial relation between the impedance plane and the incident wave.

as follows:

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} \begin{bmatrix} -H_y \\ H_x \end{bmatrix}. \tag{7}$$

In the optical axis coordinate system (uov), the surface impedance can also be defined as follows:

$$\begin{bmatrix} E_u \\ E_v \end{bmatrix} = \begin{bmatrix} Z_{uu} & 0 \\ 0 & Z_{vv} \end{bmatrix} \begin{bmatrix} -H_v \\ H_u \end{bmatrix}. \tag{8}$$

According to the coordinate transformation formula

$$\begin{bmatrix} E_u \\ E_v \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}. \tag{9}$$

By substituting (9) into (8) and comparing it with (7), the expression for surface impedance in the reference coordinate system is obtained, that is, the impedance matrix when the optical axis direction is at an arbitrary angle with respect to the reference coordinate axes is as follows:

$$\begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} = \begin{bmatrix} \cos^2\alpha Z_{uu} + \sin^2\alpha Z_{vv} & \cos\alpha\sin\alpha(Z_{uu} - Z_{vv}) \\ \cos\alpha\sin\alpha(Z_{uu} - Z_{vv}) & \cos^2\alpha Z_{vv} + \sin^2\alpha Z_{uu} \end{bmatrix}. \tag{10}$$

The reflection coefficient matrix is obtained by the impedance matrix as follows:

$$\begin{bmatrix} E_\tau \\ E_\perp \end{bmatrix} = Z_0 \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} H_\perp \\ H_\tau \end{bmatrix} \tag{11}$$

where Z_0 represent the wave impedance. From the geometric relationships in Fig. 4, it can be deduced that

$$\begin{bmatrix} E_\perp \\ E_\tau \end{bmatrix} = \begin{bmatrix} \cos\chi & \sin\chi \\ -\sin\chi & \cos\chi \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix} \tag{12}$$

$$E_\tau = E^i_{//}\cos\theta - \cos\theta E^r_{//}. \tag{13}$$

Matrix elements can be obtained from (7), (11), and (12) as follows:

$$A = \left(Z_{xx}\sin^2\chi + Z_{yy}\cos^2\chi - Z_{xy}\sin\chi\cos\chi - Z_{yx}\sin\chi\cos\chi\right)/Z_0 \tag{14a}$$

$$B = \left(Z_{xy}\sin^2\chi - Z_{yx}\cos^2\chi + Z_{xx}\sin\chi\cos\chi - Z_{yy}\sin\chi\cos\chi\right)/Z_0 \tag{14b}$$

$$C = \left(Z_{xy}\cos^2\chi - Z_{yx}\sin^2\chi - Z_{xx}\sin\chi\cos\chi + Z_{yy}\sin\chi\cos\chi\right)/Z_0 \tag{14c}$$

$$D = -\left(Z_{yy}\sin^2\chi + Z_{xx}\cos^2\chi + Z_{xy}\sin\chi\cos\chi + Z_{yx}\sin\chi\cos\chi\right)/Z_0. \tag{14d}$$

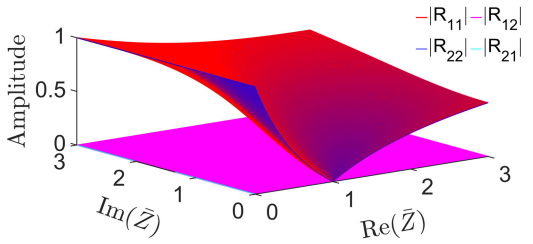


Fig. 5. Reflection coefficient variation for isotropic impedance plane.

Matrix elements of the reflection coefficient can be obtained from (5), (12), and (13) as follows:

$$R_{11} = \frac{DT^2 - \mathrm{ADT} + \mathrm{BCT} - T + A}{DT^2 + \mathrm{ADT} - \mathrm{BCT} - T - A} \tag{15a}$$

$$R_{22} = \frac{DT^2 + \mathrm{ADT} - \mathrm{BCT} + T + A}{DT^2 + \mathrm{ADT} - \mathrm{BCT} - T - A} \tag{15b}$$

$$R_{12} = \frac{-2BT}{DT^2 + \mathrm{ADT} - \mathrm{BCT} - T - A} \tag{15c}$$

$$R_{21} = \frac{-2CT}{DT^2 + \mathrm{ADT} - \mathrm{BCT} - T - A} \tag{15d}$$

in which $T = \cos\theta$. After determining the reflection coefficient matrix, the reflected EM field can be calculated using (5). Applying the plane wave excitation approximation from the PO approach, the equivalent EM current density on the anisotropic impedance surface can be derived. Finally, utilizing the Stratton–Chu far-field scattering integral, the scattering field of the scattering source, specifically the amplitude parameter A of the SC, can be obtained.

For an isotropic impedance surface, $Z_{xy} = Z_{yx} = 0$ and $Z_{uu} = Z_{vv} = Z_{xx} = Z_{yy} = Z \neq 0$. At this point, $R_{12} = R_{21} = 0$, and R_{11} and R_{22} degenerate into the common parallel and perpendicular polarization reflection coefficients.

When $\theta = 0$ and $\varphi = 0$, corresponding to a plane wave incident vertically on the impedance surface, the variation of the reflection coefficient is shown in Fig. 5, under the main range of changes in Z, which spans from $(0, 3Z_0)$.

From Fig. 5, it is evident that different impedance values exhibit varying degrees of influence on the plane's reflection characteristics. The reflection coefficient of the impedance surface at $Z = 1 + j0$ reaches its minimum value, while other impedance surfaces in the vicinity of this value display strong absorptive properties, significantly reducing the scattering intensity of incident plane waves. This characteristic can serve as a reference for stealth target design.

The impedance value $Z = 0.6 + j0.4$ is taken from Fig. 5, and the reflection coefficient amplitude is $|R_{11}| = |R_{22}| = 0.3430$. When $Z_{uu} = 0.6 + j0.4$, $\mathrm{Re}(Z_{vv}) \in (03Z_0)$, $\mathrm{Im}(Z_{vv}) \in (03Z_0)$, and $\alpha = 30°$, the variation of the reflection coefficient is shown in Fig. 6.

Compared to the isotropic impedance surface, the amplitude of the plane reflection coefficient varies significantly when the plane wave is vertically incident on the anisotropic impedance surface, which depends on the values of Z_{uu}, Z_{vv}, and α.

The impedance value that $Z_{uu} = 0.6 + j0.4$, $Z_{vv} = 1.2 + j0.2$, $\alpha = 30°$ is taken from Fig. 6, and the reflection

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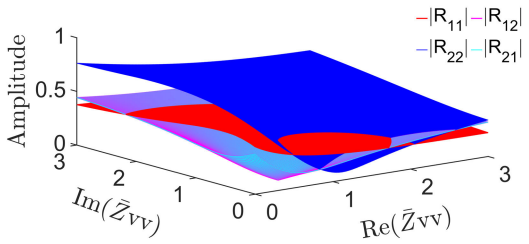


Fig. 6. Reflection coefficient variation for anisotropic impedance plane.

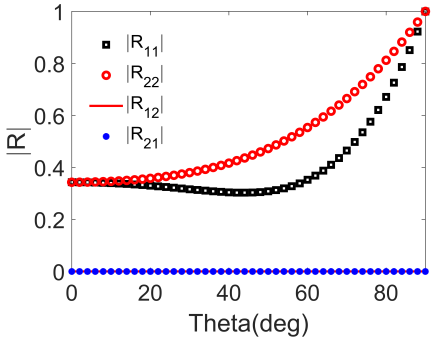


Fig. 7. Variation of the reflection coefficient in the plane of isotropic impedance with angle θ.

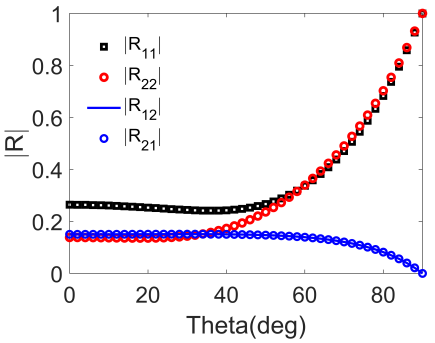


Fig. 8. Variation of the reflection coefficient in the plane of anisotropic impedance with angle θ.

coefficient amplitude is $|R_{11}| = 0.2641$, $|R_{12}| = 0.1503$, $|R_{21}| = 0.1503$, and $|R_{22}| = 0.1382$.

The variation of the reflection coefficient amplitude with an incident angle for isotropic and anisotropic planes under monochromatic plane wave irradiation is given in Figs. 7 and 8. The isotropic impedance plane parameter is $Z = 0.6 + j0.4$, and the anisotropic impedance plane parameter is $Z_{uu} = 0.6 + j0.4$, $Z_{vv} = 1.2 + j0.2$, and $\alpha = 30°$.

Examining the figures reveals that the effective attenuation of impedance surfaces to incident EM waves is primarily within the range of zenith angles from 0° to 80°, with the attenuation effect diminishing as the incident angle increases.

The EM parameters of actual materials may vary with the direction of incident waves and the wavenumber within the medium. The selection of parameters for impedance surfaces depends mainly on the application purpose. For the sake of simplifying calculations, this article assumes that the surface impedance of anisotropic materials is constant. Unless otherwise specified, the parameters of absorptive materials in Section III will be selected from two sets: $Z = 0.6 + j0.4$, $Z_{uu} = 0.6 + j0.4$, $Z_{vv} = 1.2 + j0.2$, and $\alpha = 30°$.

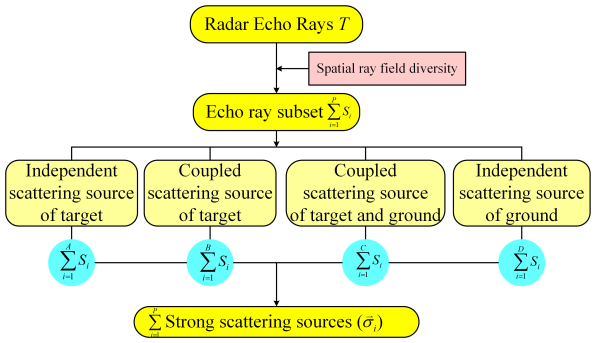


Fig. 9. Schematic of the decomposition and quantitative characterization of strong scattering sources on the ground target.

B. Statistical Analysis of Coupled SCs

The intricate and robust EM coupling scattering between the target and the rough ground exerts a substantial influence on the target's characteristics [26]. To faithfully represent the authentic scattering features of the target, it becomes imperative to account for the coupled scattering between the target and the ground, calculate the SC parameters of the coupled sources, and augment the precision of the forward model.

In high frequency, the total backscattering contribution of the ground target comes from the set of scattering sources formed by the target, the ground, and the coupling between the target and the ground, i.e., $S = [s_1, s_1, \ldots s_l, s_{l+1}, \ldots, s_{l+p}, \ldots, s_{l+p+q}]$. Its total backscattering field can be expressed as

$$E^s_{\text{total}}(f, \theta, \varphi, S) = \sum_{i=1}^{l+p+q} E^s_n(f, \theta, \varphi, S_n) \tag{16}$$

where l, p, and q represent the number of scattering sources from the ground, the target, and the target–ground coupling, respectively, i.e., the total set of scattering sources is expressed as the union of the following three subsets:

$$S_{\text{total}}(f, \theta, \varphi, s_n) = S^{\text{ground}} + S^{\text{target}} + S^{\text{coupling}}. \tag{17}$$

The schematic illustrating the decomposition and quantitative characterization of the strong scattering sources on the ground target is shown in Fig. 9.

The study employs the Monte Carlo method to numerically simulate random rough surfaces [27]. This method defines a virtual reference plane at $z = 0$ as a reference and uses $z = f(x)$ to describe the height variation characteristics of the rough surface. This function is related to the statistical parameters of the rough surface, including the root mean square height h and the correlation length l. The parameter information of the rough surface selected in this article is: dimensions $L_x = L_y = 600\lambda$ (20 m), correlation length $l_x = l_y = 120\lambda$ (4 m), root mean square height $h = 1.2\lambda$ (0.04 m), and mesh unit length 3λ (0.1 m). The schematic of the rough surface model is shown in Fig. 10.

This article integrates the high-frequency EM scattering modeling method for ground-based complex targets with the forward modeling method for SCs to achieve stable parameterization of coupled SCs. The modeling process accurately

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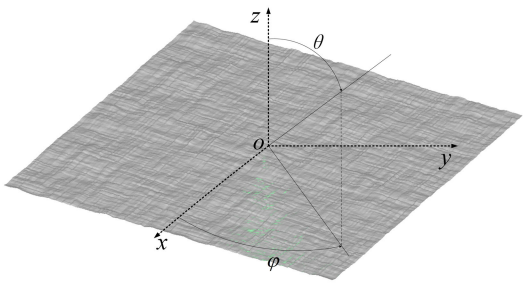


Fig. 10. Schematic of the rough surface numerical model.

incorporates and separates the contributions of complex coupling SCs between the target and the ground. To minimize the impact of introducing random rough surfaces on the calculation of coupled SC parameters, this article conducts 100 statistical implementations during the modeling process, ensuring that the results accurately reflect the target's true scattering characteristics. This provides technical support for studying EM scattering problems of coated targets with rough surfaces.

The statistical results reveal that, as the number of statistics increases, the amplitude parameters and 3-D positional coordinates of different coupled SC implementations gradually exhibit stable trends. The 3-D positional coordinates of the SCs follow the undulating pattern of random ground surfaces and tend to stabilize with increasing statistics. Additional statistics indicate a trend toward the coupling characteristics between the target and the mean plane. These statistical findings demonstrate the convergence of coupled SC parameters, with the main parameters' statistical characteristics aligning with theoretical expectations.

III. Example Analysis

This article employs a simplified tank model to illustrate the application of absorbing materials on structures with strong scattering characteristics, effectively reducing the target's RCS. The SC model for the T72 tank is forward-constructed from various observation angles to analyze the impact of coating materials and rough ground on EM scattering characteristics. Comparative analysis with actual measurements shows over 83% similarity in SAR image reconstruction. The match between peak positions in actual images and those predicted by the SC model reaches 91.95%, with a small error in matching the positions of strong SCs.

A comparative experiment is conducted between the proposed forward modeling method and the SC inverse extraction method based on simulated data, enhancing research comprehensiveness. The reconstructed SAR image exhibits 95.45% similarity in amplitude, with a calculated SC position matching rate of 96.43% and position error below 0.25 resolution units. This research thoroughly verifies the effectiveness of the SC forward modeling algorithm in reconstructing target features from both amplitude and phase information, demonstrating its robustness.

SAR target recognition relies on the structural characteristics of the targets. This article utilizes two metrics, the normalized similarity index and position matching rate, to quantitatively assess the similarity between the forward model of SCs and measurement data, considering both amplitude and phase [11], [14], [15], [28], [29]. These metrics evaluate the precision of the proposed parameterized SC model.

The normalized similarity index evaluates pixel grayscale values (amplitude) of the images, defining a feature vector representing local grayscale information. The formula calculates the similarity between feature vectors of the parameterized model-reconstructed SAR image and the measured image. Translation matching corrects potential spatial disparities. The maximum normalized correlation coefficient after translation indicates similarity. A higher value signifies closer alignment in pixel grayscale values, local numerical values, and global amplitude distribution and trends, indicating a higher level of similarity.

SC model parameters are closely related to target structural characteristics. The 3-D position parameters determine the phase of the scattering electric field, describing SC spatial phase information. This article uses the neighborhood matching algorithm to inversely extract 2-D position coordinates of the SC from the measured image. These coordinates are then compared with the forward-calculated parameters of the SC position to validate the accuracy of the phase information in the forward model.

The following provides the detailed explanations of these two metrics. Computations were conducted on a personal computer equipped with a Core i7 CPU and 8 GB of memory.

Normalized Similarity Index: The similarity between the reconstructed SAR image and the measured image feature vector of the parametric model is calculated in (18), as shown at the bottom of the next page.

In (18), $f(x, y)$ is the pixel gray value of the x row and y column of the reconstructed SAR image and $g(x, y)$ is the pixel gray value of the x row and y column of the measured image or simulated image. $\overline{f}$ is the average gray value of the reconstructed SAR image; $\overline{g}$ is the measured or the average gray value of the simulated image. Δx and Δy are the shifts in the x- and y-directions.

Position Matching Rate: The position matching rate calculation method involves using the neighborhood matching algorithm [14] to extract the 2-D position coordinates of SCs from measured images. These positions are then compared with the forward-calculated SC position parameters. Let the number of SCs output by the proposed method be M and the number extracted by the inverse method be N, and typically, M is much greater than N. The Euclidean distance between the two sets of coordinates is calculated, considering a successful match if the distance is less than two resolution units. The number of successfully matched SCs is represented as M_s, and the matching rate is calculated as shown in the following:

$$p = \frac{M_s}{M}. \tag{19}$$

A. Simplified Tank

The simplified tank is primarily composed of canonical parts with dimensions of 10 × 3.56 × 2.57 m. To accurately model

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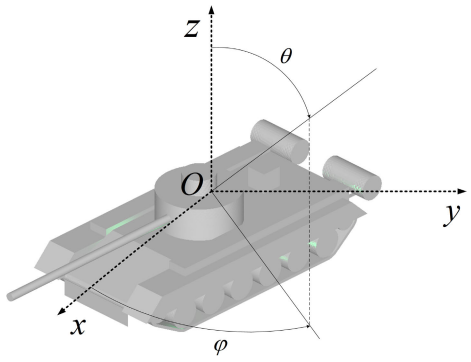
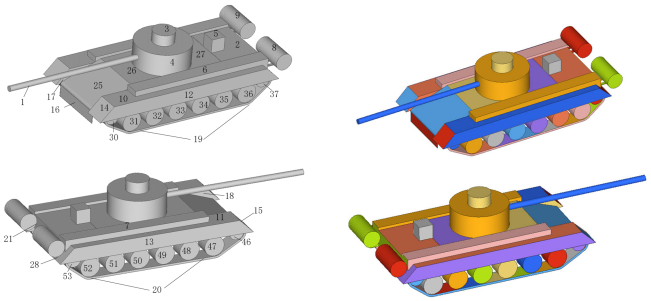


Fig. 11. Schematic of the simplified tank model.



(a) (b)

Fig. 12. Sketches of the simplified tank. (a) Components are numbered. (b) Components are highlighted by different colors.

the target, it is divided into 61 surface component parts, which are made up of around 240 000 triangular facets. A schematic of the simplified tank model can be found in Fig. 11, while Fig. 12 provides a visual representation of the simplified tank with its subcomponent parts numbered and highlighted in different colors.

The forward-calculated SC is performed at the frequency of 9 GHz, with the incidence plane wave having a zenith angle of 60° and the azimuth angle of 90°. The calculated SC parameters of the target under ideal conductive conditions are shown in Table I. The model calculation took 4 s.

At this observation angle, six distributed SCs and two local SCs are obtained through strong scattering source selection. Table I presents the predicted SCs and their associated parameters, delivering critical information such as scattering amplitude, 3-D position coordinates, and geometric structure of the target's scattering sources. The SAR image reconstruction based on this SC model under this observation posture is compared with the anechoic chamber measurements of the metal conductive model, as shown in Fig. 13. Essential image parameters are available in Table II.

The eight strong scattering sources of the simplified tank target are clearly presented in the reconstructed SAR image, and the corresponding relationship between the eight SCs and the geometric components of the tank target is shown in Fig. 13 (right). Among them, SC 1 is generated by the reflection of the dihedral angle formed between the left wheel cover and the left track plate, and this SC has the largest amplitude and is of a distributed type. SC 2 is generated by the reflection of the dihedral angle formed between the cube and the roof, while SC 3 is generated by the first-order reflection of the gun barrel. SCs 4–6 are generated by the reflection between the right track guard and the roof, while SCs 7 and 8 are generated by the double reflection between the left track guard and the turret.

The above comparative target is a metal conductive model. How the scattering properties of the target are modified when the target is locally coated with an absorbing material will be explained in the following through models and variations of the scattering properties. The scattering characteristics of the target are primarily determined by its strong scattering sources, as listed in Table I. Therefore, by coating the surfaces of these components, the scattering characteristics of the target can be controlled. In this article, the coating used is a thickness-free impedance surface with an isotropic impedance parameter $Z = 0.6 + j0.4$ and an anisotropic impedance parameter $Z_{uu} = 0.6 + j0.4$, $Z_{vv} = 1.2 + j0.2$, and $\alpha = 30°$. As revealed in the section of impedance reflection characteristics analysis, it is expected that the two types of impedance surfaces will cause nearly half of the attenuation effect on the target scattering under this observation attitude. Fig. 14 shows a schematic of the locally coated target, with the red highlighted area representing the coated part. The variations in the scattering amplitudes before and after coating the strong scattering sources of the target are presented in Table III.

The data in Table III indicate that the amplitude of SC 1 is reduced by 31.42% and 44.65%, respectively, to 29.55 and 23.85 dB. The amplitudes of SCs 2 and 3 are also significantly attenuated, which is consistent with expectations. The reconstruction results show that the influence of the absorptive coating on the SCs mainly manifests in changes in the amplitude of the SCs. Since the target's geometric structure did not change significantly before and after coating, the 3-D position coordinates, length parameters, and other parameters of the SCs remained basically unchanged. Furthermore, the attenuation effect of different material parameters on target scattering is not the same, depending on the reflective properties of the coating material, as discussed earlier. Figs. 15 and 16 present the reconstructed SAR images of the target SC model calculated under two coating material scenarios compared with the metal conductive model.

The amplitude parameters of target SCs exhibit significant transformations before and after coating. Beyond the substantial amplitude reduction shown in Table III, the SAR images of the coated target show diminishing brightness in strong scattering structures such as the cylindrical shape of the gun

$$\mathrm{Cor} = \max\left\{ \frac{\sum_x \sum_y \left[f(x,y) - \overline{f}\right]\left[g(x-\Delta x, y-\Delta y) - \overline{g}\right]}{\left\{\sum_x \sum_y \left[f(x,y) - \overline{f}\right]^2 \sum_x \sum_y \left[g(x-\Delta x, y-\Delta y) - \overline{g}\right]^2\right\}^{1/2}} \right\} \tag{18}$$

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TABLE I

FORWARD MODELING RESULTS OF STRONG SCS FROM SIMPLIFIED TANK (PEC) AT $f = 9$ GHz, $\theta = 60°$, AND $\varphi = 90°$

No.	Index series	Equivalent canonical geometries	Scattering source	RCS (dBsm)	α	L (m)	3-D positions (m)
1	6-10	Front dihedral	Double reflection between the left wheel cover and left track shield	43.09	1	4.98	(-0.55,1.5,1.27)
2	2-5	Middle dihedral	Double reflection between the cube and vehicle roof	29.38	1	0.48	(-2.05,0.25,1.37)
3	1	Cylindrical surface	Single reflection from the barrel	27.02	0.5	5.00	(3.49,0.09,1.82)
4	2-7	Back dihedral # 1	Double reflection between the right track shield # 1 and vehicle roof	24.75	1	0.93	(-2.78,-1.00,1.37)
5	7-27	Back dihedral # 2	Double reflection between the right track shield # 2 and vehicle roof	23.39	1	0.79	(-1.40,-1.00,1.37)
6	7-26	Back dihedral # 3	Double reflection between the right track shield # 3 and vehicle roof	17.61	1	0.72	(1.39,-1.00,1.37)
7	4-6	Top-hat # 1	Double reflection between the left track shield and turret	16.76	0.5	0.00	(0.00,1.00,1.47)
8	3-4	Top-hat # 2	Double reflection between the upper and lower layers of the turret	13.01	0.5	0.00	(0.00,0.39,2.17)
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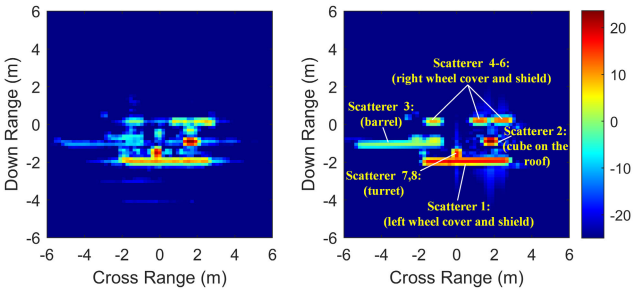
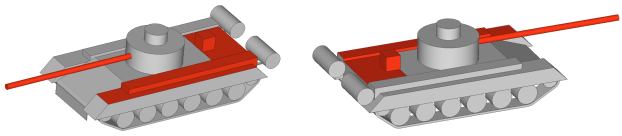


Fig. 13. Comparison between actual (left, metallic conducting model) and reconstructed (right, PEC) SAR images for simplified tank at $\theta = 60°$ and $\varphi = 90°$. The similarity is 87.32%.

TABLE II

IMAGING PARAMETERS OF SIMPLIFIED TANK

Polarization	HH	Zenith angle	θ
Center Frequency	9GHz	Aspect angle	φ
Bandwidth	1GHz	Angular Spacing	0.078°
Frequency Step	10MHz	Synthesis Aperture	7.78°
Range Resolution	0.15m	Azimuth Resolution	0.15m



(a) (b)

Fig. 14. Schematic of target local coating. (a) Frontal oblique view. (b) Rear oblique view.

TABLE III

RCS VARIATION OF SCS UNDER DIFFERENT COATING CASES

No.	Index series	RCS(dBsm)		
		PEC	$Z = 0.6 + j0.4$	$Z_{uu} = 0.6 + j0.4$ $Z_{vv} = 1.2 + j0.2$ $\alpha = 30°$
1	6-10	43.09	29.55	23.85
2	2-5	29.38	15.84	10.06
3	1	27.02	17.16	12.36

barrel, the dihedral angle of the wheel cover and track plate, and the cube and vehicle roof dihedral angle. These structures transition into weaker scattering sources of the target, with formerly weaker scattering sources now emerging as strong scattering sources. The similarity between the reconstructed

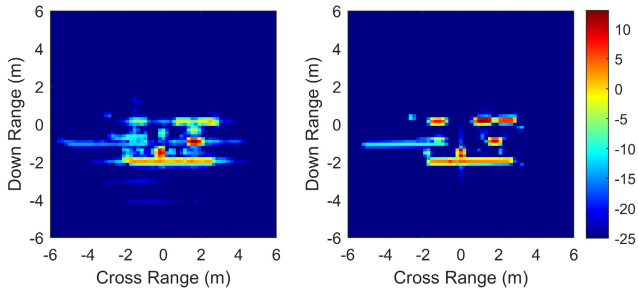


Fig. 15. Comparison between actual (left, metallic conducting model) and reconstructed (right, locally coated with isotropic impedance) SAR images for simplified tank at $\theta = 60°$ and $\varphi = 90°$. The similarity is 70.45%.

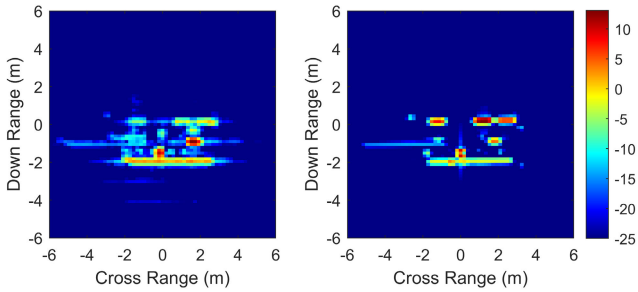


Fig. 16. Comparison between actual (left, metallic conducting model) and reconstructed (right, locally coated with anisotropic impedance) SAR images for simplified tank at $\theta = 60°$ and $\varphi = 90°$. The similarity is 56.20%.

SAR images for the extended model under the two coating conditions and the metallic conducting target is 70.45% and 56.20%, respectively, signifying a decline in similarity. This illustrates that the application of absorbing materials alters the distribution of strong scattering sources and the EM characteristics of the target.

This article establishes a forward model for SCs of coated targets, establishing a mapping relationship between SCs and local scattering structures of the target. It directly reflects parameter values associated with the target's local geometry and coating material properties. This model provides efficient guidance for rapid estimation and analysis of radar target characteristics in complex real-world environments, supporting applications in remote sensing target recognition and related fields.

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Fig. 17. Authentic photograph of the T72 main battle tank.

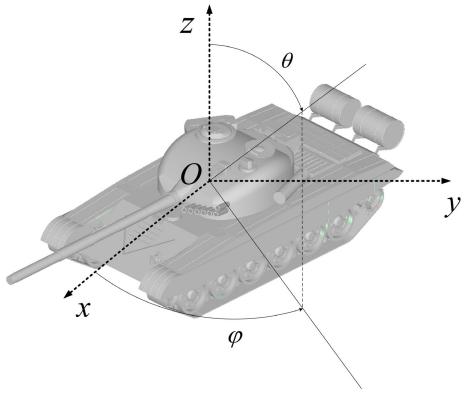


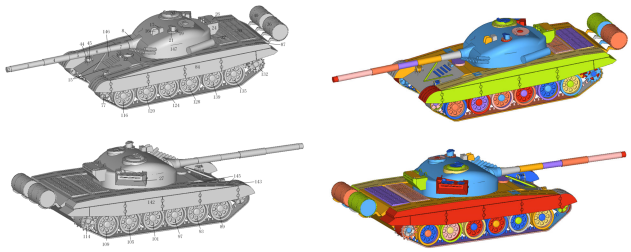
Fig. 18. Schematic of the T72 tank model.

B. T72

Liu et al. [11] verified the effectiveness of the attribute SC parameterization forward automatic modeling method in replicating the EM scattering features of intricate targets, such as the T72 tank. However, in practical and complex environments, targets are often not constructed from PEC materials. Targets made of highly conductive alloys may feature surfaces coated with nonmetallic dielectric materials, such as absorbent coatings or rust-proof paint, introducing "surface" loss effects, among other factors. In this study, the non-PEC material surface of the T72 target, resulting from aging and rusting over time, is simulated using material dielectric parameters. A thorough examination and analysis of the agreement between the reconstructed EM characteristics of the coated T72 target SC model and measurement results are conducted. An authentic photograph of the T72 tank is presented in Fig. 17.

The T72 main battle tank has dimensions of 7.54 × 2.52 × 1.91 m. To achieve the accuracy of the SC model given the complexity of the target structure, the target computer-aided design (CAD) model was decomposed into 147 surface regions, consisting of approximately 530 000 facets. The schematic of the model and the partitioning diagram are shown in Figs. 18 and 19, respectively.

The extraction of SCs is conducted at a frequency of 9.6 GHz, with the incidence plane wave having an elevation angle of 73° and an azimuth angle of 60°. The calculated results of SC parameters for the ideal conducting target are shown in Table IV. On a computer with Core i7 CPU and 8-GB RAM, obtaining the parametric models requires 20 s.



(a) (b)

Fig. 19. Sketches of T72 tank. (a) Components are numbered. (b) Components are highlighted by different colors.

Compared to the simplified tank, the surface of the T72 tank contains mostly atypical scattering structures, and the amplitude of the scattering sources is significantly reduced. To improve the accuracy of the SC model, the target CAD model is disassembled into 147 surface components. Table IV lists the first eight strong SCs in order of amplitude, with the index series column indicating the area serial number of different parts of the tank, such as the left wheel cover (87), turret (147), and outer surface of the left wheel (116–140). SC 1 is a result of reflection between the turret and left wheel cover, while SC 2 is from the primary scattering of the turret. SCs 3–8 are generated from double scattering of the wheel hubs and wheel frame of the multiple supporting wheels on the left side of the tank. The SAR images of the target in this attitude are reconstructed, and the scattering sources corresponding to the SCs are marked on the images and compared with the actual images of the MSTAR dataset. The results are shown in Fig. 20, and Table V presents the image parameters.

It is noteworthy that, given the challenge of acquiring precise geometry data for the actual T72 tank target, the geometric model utilized in this article is not entirely accurate. The inaccuracy of the target CAD model remains the primary limiting factor in achieving consistency between the reconstructed SC model of the target characteristics and the measured data.

The gun turret and left wheel cover, as strong scattering sources for PEC target, are significantly brighter in the reconstructed SAR image than other scattering sources. However, the measured data at these locations are significantly smaller than the reconstructed characteristic data for PEC target. The SCs are concentrated around geometric features such as the turret and left wheel cover, representing strong scattering sources. The distribution of SCs on the PEC target differs from the actual scenario. Considering that the target material is not composed of ideal PECs, the surface of target made of highly conductive alloys may be coated with nonmetallic dielectric materials, such as absorptive coatings or rust-proof paint or even locally coated with absorbing materials to achieve the effect of "surface" loss and so on. Therefore, it should be considered whether the difference between the reconstructed PEC model target characteristics and the measured data is related to the reduced conductivity of the target surface after "painting." By selecting appropriate coating medium parameters and designing the coated target SC model, target characteristic data

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TABLE IV

FORWARD MODELING RESULTS OF STRONG SCS FROM T72 TANK (PEC) AT $f = 9.6$ GHz, $\theta = 73°$, AND $\varphi = 60°$

No.	Index series	Equivalent canonical geometries	Scattering source	RCS (dBsm)	α	L (m)	3-D positions (m)
1	87-147	Cylinder above the flat plane	Double reflection between the left wheel cover and turret	8.55	0.5	0.05	(0.69,0.78,1.16)
2	147	Single curve surface	Single reflection from turret	0.01	0.5	0.10	(0.59,0.80,1.49)
3	116-117	Flat and cylindrical surfaces	Single reflection from the barrel	-2.32	0.5	0.02	(1.64,1.06,0.32)
4	128-129	Flat and cylindrical surfaces	track roller	-2.36	0.5	0.02	(-0.20,1.06,0.32)
5	135-136	Flat and cylindrical surfaces	track roller	-2.51	0.5	0.02	(-1.44,1.06,0.32)
6	120-121	Flat and cylindrical surfaces	track roller	-2.74	0.5	0.02	(1.00,1.06,0.33)
7	139-140	Flat and cylindrical surfaces	track roller	-3.73	0.5	0.02	(-0.84,1.06,0.33)
8	124-125	Flat and cylindrical surfaces	track roller	-4.99	0.5	0.02	(0.40,1.06,0.32)
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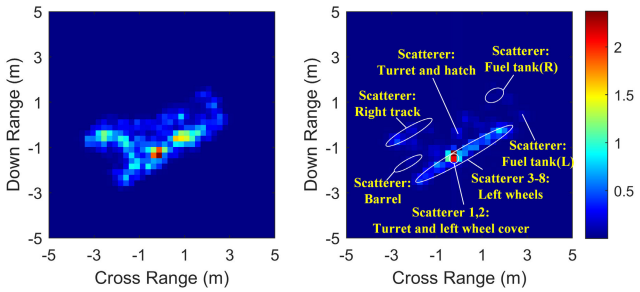


Fig. 20. Comparison between actual (left) and reconstructed (right, PEC) SAR images for T72 at $\theta = 73°$ and $\varphi = 60°$. The similarity is 75.87%.

TABLE V

IMAGING PARAMETERS OF SIMPLIFIED TANK

Polarization	HH	Zenith angle	θ
Center Frequency	9.6GHz	Aspect angle	φ
Bandwidth	0.6GHz	Angular Spacing	0.097°
Frequency Step	15MHz	Synthesis Aperture	3.87°
Range Resolution	0.25m	Azimuth Resolution	0.25m

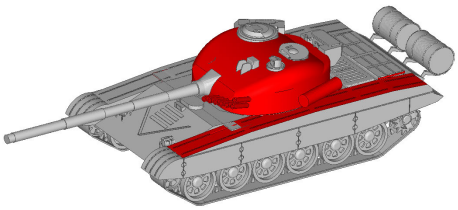


Fig. 21. Schematic of the local coating case for the T72 tank target, at $\theta = 73°$ and $\varphi = 60°$.

TABLE VI

RCS VARIATION OF SCS UNDER DIFFERENT COATING CASES

No.	Index series	RCS(dBsm)		
		PEC	$Z = 0.6 + j0.4$	$Z_{uu} = 0.6 + j0.4$ $Z_{vv} = 1.2 + j0.2$ $\alpha = 30°$
1	87-147	8.55	-3.73	-12.47
2	147	0.01	-9.11	-13.64

can be obtained, which can then be compared and analyzed with the measured characteristic data to examine the utility and rationality of the coated target SC model.

Fig. 21 shows a schematic of the local coating on the T72 tank target, with the coated parts highlighted in red. Table VI provides the RCS variants of the two strong scattering sources, the turret, and the left wheel cover, for different cases of coating. The isotropic impedance parameter is $Z = 0.6 + j0.4$, and the anisotropic impedance parameter is $Z_{uu} = 0.6 + j0.4$, $Z_{vv} = 1.2 + j0.2$, and $\alpha = 30°$. Fig. 22 shows the variation in SC intensity created by different coating materials on specific target components. In comparison to the PEC case, as the coating material's absorptive characteristics are enhanced, the SC intensity rapidly decreases. This demonstrates a change in SC magnitude. Figs. 23 and 24 show the reconstructed SAR images of the SC model for two coated targets compared with the MSTAR measured data.

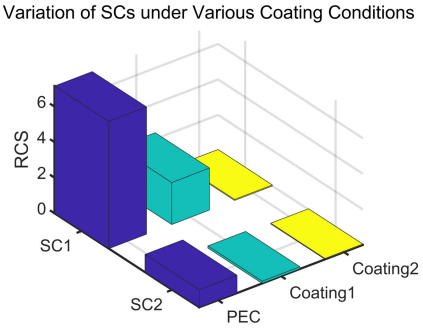


Fig. 22. Variation in SCs intensity created by specific target components under different coating material conditions at $\theta = 73°$ and $\varphi = 60°$.

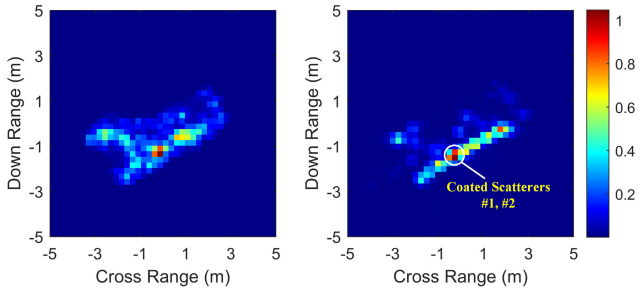


Fig. 23. Comparison between actual (left) and reconstructed (right, locally coated with isotropic impedance) SAR images for T72 at $\theta = 73°$ and $\varphi = 60°$. The similarity is 79.59%.

Table VI and Fig. 22 show that the application of absorbing materials on the surface of the target results in a significant attenuation of the scattering amplitude of the coated scattering source. The scattering intensity of SC 1 decreases from 8.55 to -3.73 dB or even -12.47 dB. It can also be observed from the reconstructed SAR images that the brightness of the strong scattering source of the coated target significantly decreases, and the weaker scattering sources become dominant. The distribution of strong scattering sources of the target changes.

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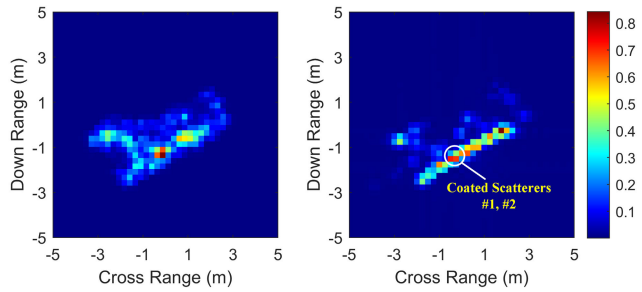


Fig. 24. Comparison between actual (left) and reconstructed (right, locally coated with anisotropic impedance) SAR images for T72 at $\theta = 73°$ and $\varphi = 60°$. The similarity is 77.1%.

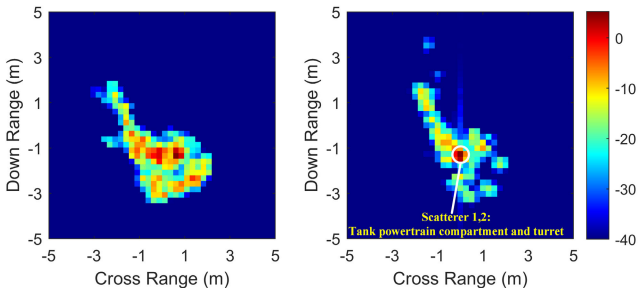


Fig. 25. Comparison between actual (left) and reconstructed (right, PEC) SAR images for T72 at $\theta = 73°$ and $\varphi = 160°$. The similarity is 78.48%.

TABLE VII

FORWARD MODELING RESULTS OF STRONG SCS FROM T72 TANK (PEC) AT $f = 9.6$ GHz, $\theta = 73°$, AND $\varphi = 160°$

No.	Index series	RCS (dBsm)	α	L (m)	3-D positions (m)
1	32-147	4.94	0	0.037	(-0.93,0.39,1.20)
2	30-147	-3.49	0	0.032	(-0.94,0.36,1.19)
3	147	-6.97	0	0	(-0.93,0.39,1.47)
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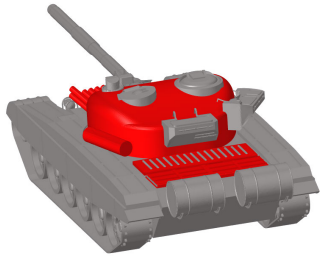


Fig. 26. Schematic of the local coating case for the T72 tank target, at $\theta = 73°$ and $\varphi = 160°$.

TABLE VIII

RCS VARIATION OF SCS UNDER COATING

No.	Index series	RCS(dBsm)	
		PEC	$Z = 0.6 + j0.4$
1	32-147	4.94	-6.93
2	30-147	-3.49	-15.07
3	30-147	-6.97	-16.43

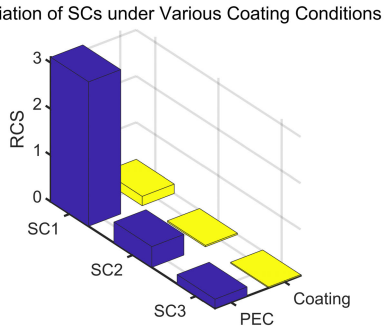


Fig. 27. Variation in SCs intensity created by specific target components under different coating material conditions at $\theta = 73°$ and $\varphi = 160°$.

The similarity between the reconstructed SAR images of the two types of coated target and the MSTAR dataset is 79.59% and 77.1%, respectively, both of which are higher than that of the PEC target. In actual scenarios, the antirust paint or absorbing material on the surface of the T72 tank is usually a thin coating, and the geometry of the target does not change significantly before and after coating. Therefore, the 3-D position coordinates and length parameters of the SC remain consistent with the PEC case. The forward modeling process of the coated T72 target accurately establishes a mapping relationship between the SC and the local scattering structure of the target. The scattering characteristics of the target reconstructed using the two sets of coated target SC models are closer to the "real" radar characteristics of the target than the PEC case, which verifies the utility and rationality of the coated target SC model.

With the azimuth angle changed to 160°, Table VII presents the top 3 strong SC-related parameters of T72 tank (PEC) sorted by amplitude under this observation orientation.

Index series 30 and 32 refer to the surface of the power cabin, while index series 147 corresponds to the turret. SCs 1 and 2 originate from the coupling scattering of the power cabin and the turret, respectively, while SC 3 originates from the first scattering of the turret. The reconstructed SAR image of the target in this posture is shown in Fig. 25, with the scattering sources corresponding to the SCs marked on the image and compared with the measured image.

The reconstructed SAR image of the target under the PEC condition achieves a 78.48% similarity with the measured image. Notably, the powertrain compartment and turret surfaces, prominent scattering sources on the target, exhibit significantly higher brightness in the image compared to other scattering sources. However, the measured data in those areas are lower than the reconstruction results. Consequently, a detailed analysis of alterations in the target's reconstructed scattering characteristics following the application of absorbing materials to these strong scattering sources is imperative.

Fig. 26 shows the local coating on the T72 tank target, with the coated parts highlighted in red. Table VIII presents the RCS variations of the power cabin and turret surfaces of the tank after being coated with absorbing materials. The impedance parameter is $Z = 0.6 + j0.4$. Fig. 27 shows the 3-D visualization of RCS magnitude changes of SCs before and after coating using histogram. Fig. 28 shows the comparison between the reconstructed SAR image of the coated target's SCs and the measured data.

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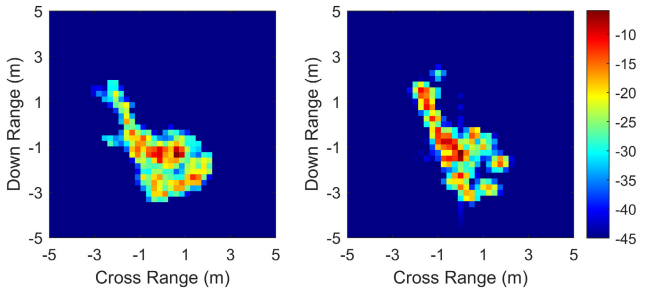


Fig. 28. Comparison between actual (left) and reconstructed (right, locally coated with media) SAR images for T72 at $\theta = 73°$ and $\varphi = 160°$. The similarity is 79.30%.

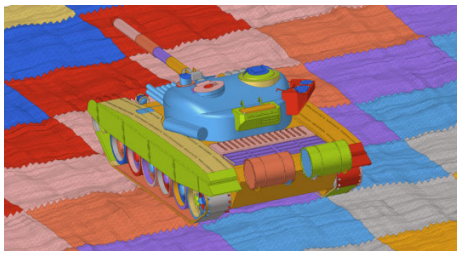


Fig. 29. Observed attitude of the T72 tank on a rough surface at $\theta = 73°$ and $\varphi = 160°$.

The target coating enhances the relative amplitude distribution and overall SC distribution compared to PEC cases, aligning with the measured data. However, an evident disparity remains: the reconstructed SAR images in free space, whether coated or not, exhibit a pronounced absence of scattering contribution in the local range of the lower right corner, as opposed to the measured images. Therefore, it is crucial to investigate the impact of coupling SCs arising from the interaction between the target and rough surfaces on the target's scattering characteristics. This consideration becomes particularly significant since the T72 tank testing environment encompasses flat and rough ground.

C. T72 Tank on Rough Surface

The observation posture of the tank target on the ground is shown in Fig. 29. The stable SC parameters formed by the coupling scattering between the rough surface and the target are statistically obtained through the Monte Carlo method. Fig. 30 shows the statistical results of the amplitude and 3-D position parameters of SCs formed by the coupling between several typical parts (numbered 2, 44, and 129) on the target and the rough surface. With the increase of statistical times, the continuous fluctuations of amplitude parameters and 3-D position coordinates of SCs with different implementations gradually show a stable trend. The fluctuation of the 3-D position coordinates of the SC conforms to the roughness rule of random ground and gradually tends to be stable with the increase of statistical times. Further statistical results converge to the coupling feature between the target and the mean plane of the ground. Table IX lists the statistical results of SCs formed by the coupling between the target and the ground environment.

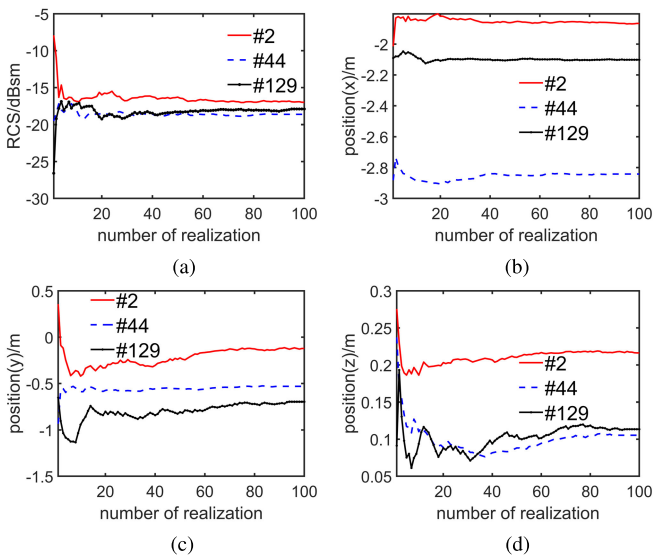


Fig. 30. Statistical results of coupled SCs. (a) Amplitude. (b) Position (x). (c) Position (y). (d) Position (z).

TABLE IX
STABLE COUPLING SC PARAMETERS

No.	Index series	RCS (dBsm)	3-D positions (m)
1	1-197	-16.98	(-1.87,-0.12,0.22)
2	36-186	-18.11	(-2.78,0.13,0.3)
3	40-186	-16.11	(-2.66,-0.42,0.3)
4	35-176	-19.46	(-0.52,0.33,0.0)
5	84-187	-18.77	(-1.59,0.9,0.08)
6	84-176	-22.76	(-2.07,0.83,0.0)
7	59-198	-21.2	(-1.39,0.99,0.1)
8	133-198	-24.56	(-1.78,0.98,0.14)
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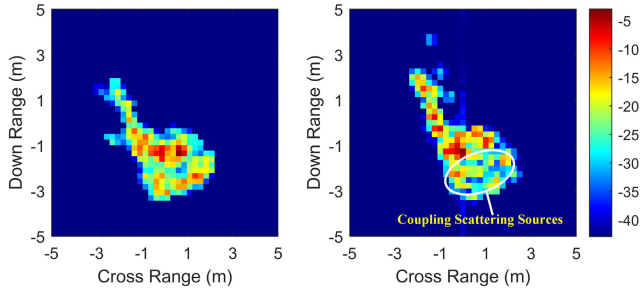


Fig. 31. Comparison between actual (left) and reconstructed (right, statistical characteristics) SAR images for the T72 tank at $\theta = 73°$ and $\varphi = 160°$. The statistical similarity is 83.53%.

Index series 148–267 correspond to the rough ground areas. Index series 1 represents the inclined plate behind the tank, and index series 36 and 40 pertain to the outer surfaces of the left and right fuel tanks of the tank. Index series 84 designates the left wheel cover of the tank, and index series 59 and 133 denote the outer surfaces of the left driving wheels of the tank.

Fig. 31 shows the comparison between the statistically reconstructed SAR image and the measured data of the T72 tank target on the rough ground.

In Fig. 32, the SCs formed by the coupling between the target and the environment are marked with black "×" symbols

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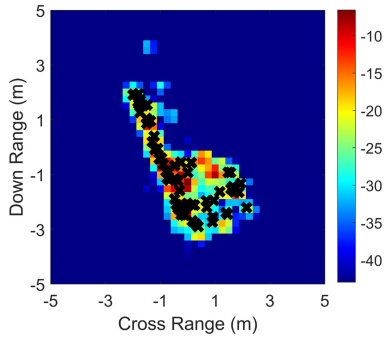


Fig. 32. Projection of coupled SCs on the SAR image.

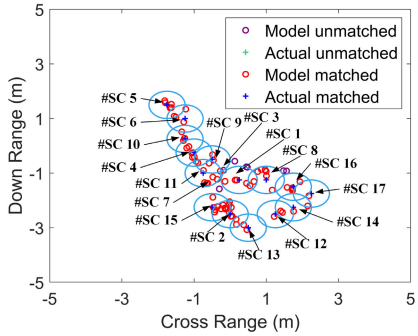


Fig. 33. Matching results of the measured image's inverse-extracted SC positions and the forward-calculated SC positions for T72 at $\theta = 73°$ and $\varphi = 160°$, and the position matching rate is 91.95%.

on the SAR image to reflect the distribution of coupling SCs in the SAR image.

In Fig. 32, coupling SCs are distributed around the target, mainly concentrated on the side closer to the radar, consistent with the real scenario. Under this observation angle, coupling SCs generated by the interactions between the tank's rear sloped plate, fuel tank, side wheel covers, and the rough surface are evenly distributed in the lower right corner of the target's 2-D coordinates. This effectively compensates for the missing scattering sources in the reconstructed SAR image of the target in that range compared to actual measurements.

Fig. 33 shows the overall SC position matching results. The distribution of SC positions from the forward model closely aligns with actual measurements, resulting in an overall position matching rate of 91.95%. The results for matching the positions of strong SCs, constructed by the forward model, are listed in Table X, with position matching errors well within one resolution unit. These comparative results provide clear evidence of the forward model's accurate reconstruction of target feature data, particularly in the phase domain, showing significant similarity to the measured data. Together with the earlier discussed normalized similarity index, this research validates the model's accuracy from both amplitude and phase aspects.

D. Comparison of SC Forward-Calculated and Inverse Extraction Methods

This section conducts a comparative experiment between the SC inverse extraction method, based on simulated data, and the forward modeling approach proposed in this article, aiming to enhance the research's comprehensiveness. To ensure comparability and credibility in the comparative experiment, both methods employ the same geometric model. The experiment uses the neighborhood matching algorithm to inversely extract the 2-D position coordinates of the SCs from the simulated SAR image of the target. These coordinates are then compared with the position parameters obtained from the forward calculation of SCs.

Continuing with the T72 tank geometric model utilized in Section III-B, the results of forward modeling for strong SCs at an observation angle of $\theta = 73°, \varphi = 300°$ are shown in Table XI. The forward model provides information about the corresponding scattering sources and relevant attribute parameters of the strong SCs.

Table XI lists the top 15 strong SCs sorted by amplitude. The index series indicates region serial numbers for different parts of the tank, such as the turret (147), the front V-shaped mudguard (145), the support wheels inside the left track of the tank (89–102), and the side armor of the tank. SC 1 is due to the coupled scattering between the mudguard and the turret, SC 2 arises from the single scattering of the turret, and SCs 3–7 are from the inner and outer wall scattering of multiple support wheels on the side of the tank. SCs 8–11 are contributed by the scattering from the front armor. Coupled scattering between the front mudguard and side armor generates SC 12, while SCs 13–15 result from the coupled scattering between the turret and occasional components on the turret.

The comparison of amplitude similarity between the forward model-reconstructed SAR image and the simulated image is shown in Fig. 34, with an image similarity of 95.45%. The overall results of SC positions are presented in Fig. 35. The distribution of SC positions from the forward model closely matches the results obtained from reverse extraction of simulated data, with an overall position matching rate of 96.43%. The matching results for the strong SCs of the target are provided in Table XII, with position matching errors within 0.25 resolution units. The forward-calculated positions of SCs closely match the peak values in the simulated image. These two evaluation metrics confirm the accuracy of the model in reconstructing target characteristics, validating the correctness and effectiveness of the proposed forward modeling method for SCs.

It is essential to emphasize that the SC inverse extraction method is data-driven and focused on feature inversion. It can solely extract 2-D position coordinates from image data, leading to limitations. When multiple 3-D SCs overlap and project onto the imaging plane, the inverse method can retrieve only one SC, resulting in an underestimated count of extracted SCs. Similarly, since the inverse method can only extract 2-D position information of SCs, obtaining the 3-D positions requires a comprehensive analysis of many poses across the entire scope. Whether it is possible to obtain reasonable 3-D positions remains inconclusive without reliable data support.

In contrast, the SC forward modeling method introduced in this article initiates from the target's CAD model, employing spatial ray tracing and diversity techniques for forward

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TABLE X

RCS VARIATION OF SCS UNDER COATING

No.	Scattering center forward modeling			Measured image's inverse-extracted SC position	Position matching error (resolution unit)
	Index series	Scattering source	Forward-calculated SC position		
1	32-147	Power cabin and turret	(0.138,-1.253)	(0.25,-1.25)	**0.45**
2	36-176	Fuel tank and rough ground	(-0.025,-2.450)	(0,-2.5)	**0.20**
3	128-129	Track rollers #1	(-1.028,-0.251)	(-1,-0.25)	**0.10**
4	116-117	Track rollers #2	(-1.629,1.402)	(-1.75,1.50)	**0.48**
5	120-121	Track rollers #3	(-1.285,0.876)	(-1.25,1.00)	**0.50**
6	27-30	Main fuel tank	(1.003,-1.100)	(1,-1.25)	**0.60**
7	35-187	Turret and rough ground	(-0.465,-0.524)	(-0.5,-0.5)	**0.14**
8	124-125	Track rollers #4	(-1.268,0.271)	(-1.25,0.25)	**0.08**
9	139-140	Track rollers #5	(-0.741,-0.9)	(-0.75,-1)	**0.40**
10	40	Right fuel tank	(1.220,-2.588)	(1.25,-2.5)	**0.35**

TABLE XI

FORWARD MODELING RESULTS OF STRONG SCS FROM T72 TANK AT $f = 9.6$ GHz, $\theta = 73°$, AND $\varphi = 300°$

No.	Index series	Scattering source	RCS (dBsm)	α	L (m)	3-D positions (m)
1	145-147	Turret and V-shaped splash guard	8.02	1	0.05	(0.69,-0.77,1.16)
2	147	Turret	-2.31	0.5	0.85	(0.69,-0.57,1.19)
3	93-94	Track roller #1	-2.37	0.5	0.02	(1.01,-1.12,0.32)
4	109-100	Track roller #2	-2.68	0.5	0.02	(-1.32,-1.27,0.33)
5	105-106	Track roller #3	-2.75	0.5	0.02	(-0.73,-1.23,0.32)
6	101-102	Track roller #4	-3.77	0.5	0.02	(-0.15,-1.21,0.33)
7	89-90	Track roller #5	-3.82	0.5	0.01	(1.94,-1.18,0.33)
8	85	Sloping left front armor	-8.91	0.5	0	(2.94,0.78,0.86)
9	143	Sloping right front armor	-9.65	0.5	0	(2.94,-1.34,0.88)
10	143-144	Right armor	-11.46	0	0	(2.97,-1.34,0.80)
11	85-86	Left armor	-11.88	0	0	(2.98,0.82,0.79)
12	86-146	Armor and V-shaped splash guard	-13.29	1	0	(1.99,0.85,0.83)
13	27-147	Turret and Main fuel tank	-13.40	1	0	(-0.75,-0.16,1.62)
14	21-147	Periscope and turret	-14.18	0.5	0	(0.49,0.29,1.73)
15	19-147	Hatch cover and turret	-14.22	0.5	0	(0.44,0.32,1.73)
...	...	...	...	...	...	...

TABLE XII

MATCHING RESULTS OF STRONG SC POSITIONS

No.	Scattering center forward modeling			Simulated image's inverse-extracted SC position	Position matching error (resolution unit)
	Index series	Scattering source	Forward-calculated SC position		
1	93-94	Track roller #1	(0,-1.5)	(0,-1.5)	**0.00**
2	109-110	Track roller #2	(-2,-0.51)	(-2,-0.5)	**0.04**
3	105-106	Track roller #3	(-1.49,-0.76)	(-1.5,-0.75)	**0.04**
4	101-102	Track roller #4	(-1,-1.03)	(-1,-1)	**0.12**
5	89-90	Track roller #5	(0.75,-2)	(0.75,-2)	**0.00**
6	85	Sloping left front armor	(2.51,-1)	(2.5,-1)	**0.04**
7	143	Sloping right front armor	(1.49,-2.77)	(1.5,-2.75)	**0.08**
8	143-144	Right armor	(1.52,-2.76)	(1.5,-2.75)	**0.08**
9	85-86	Left armor	(2.56,-0.98)	(2.5,-1)	**0.24**
10	86-146	Armor and V-shaped splash guard	(1.76,-0.49)	(1.75,-0.5)	**0.04**

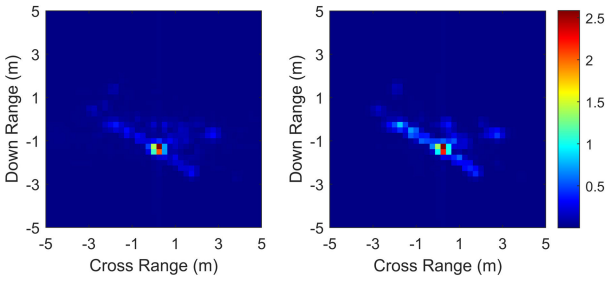


Fig. 34. Comparison between simulated (left) and reconstructed (right) SAR images for T72 at $\theta = 73°$ and $\varphi = 300°$. The similarity is 95.45%.

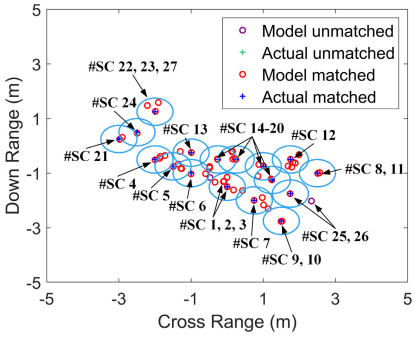


Fig. 35. Matching results of the simulated image's inverse-extracted SC positions and the forward-calculated SC.

calculation and quantification of attributes, including the accurate 3-D coordinates of SC models. This method follows a "cause-to-effect" forward approach, offering clear correspondences with local target structures. It carries genuine physical significance and eliminates the need for complex iterative optimization processes. Leveraging the forward model's advantages, it provides a detailed deconstruction of target EM scattering, facilitating in-depth exploration of target feature matching information.

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IV. Conclusion

This research is dedicated to applications in remote sensing and target identification, presenting a pioneering study on forward modeling of SCs for coated targets on rough ground. It establishes the overall framework for SC forward modeling, providing a detailed calculation process for SC parameter assignment. In contrast to traditional inverse methods, this approach follows a "cause-to-effect" forward path, offering a clear physical interpretation.

Using the T72 tank from the MSTAR database as an example, the forward modeling approach constructs the target's SC model from multiple viewing angles, conducting an in-depth analysis of the effects of coating materials and rough ground on the target's EM scattering characteristics. Comparative analysis with measured results indicates that the model achieves a high degree of similarity in SAR image reconstruction, surpassing 83%. The matching rate between peak points in the measured image and SC model positions reaches 91.95%, affirming the accuracy and effectiveness of this approach. In comparison to inverse extraction methods, the SC forward modeling method directly links SCs to the target's local structure, eliminating the need for complex iterative optimization processes. It excels in providing a clear physical deconstruction of the target's EM scattering, facilitating deep exploration of target feature matching information.

Moreover, this research extends the application of the forward modeling method from ideal conductive targets in free space to dielectric-coated targets in realistic rough ground environments. The model parameters encompass local target geometry, coating material properties, and stable characteristics of coupling SCs. This enhancement enables the model to more accurately represent real-world EM scenarios. The research broadens the scope of forward modeling applications and offers a cost-effective technique for rapid estimation and analysis of radar characteristics in complex real-world environments, as well as target detection and identification.

In the future, this method holds potential for applications in RCS reduction, false scattering elimination, camouflaged target identification, and parameterized modeling of noncooperative targets, providing valuable tools for various target remote sensing applications.

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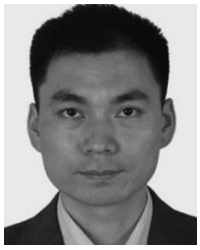
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